

BRICEN PEINADO

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EDUCATION

Binghamton University - Binghamton, NY **2026 - Present**
Bachelor of Science in Computer Science | Junior
Suffolk County Community College: Associate of Science in Computer Science | GPA: 3.4 | Coursework: Data Structures & Algorithms, OOP, Computer Architecture, Discrete Math, Linear Algebra

RESEARCH EXPERIENCE

Undergraduate AI/ML Researcher **2026 - Present**
Meta-GEPA: Self-Referential Reflection Tuning for Resolving Conversational Ambiguity

- Designing a prompt-evolution framework that uses LLM evaluator feedback and self-referential reflection to improve ambiguity resolution without model fine-tuning or clarification turns.
- Building a 20-case benchmark and heuristic evaluation framework spanning lexical, syntactic, referential, semantic, and pragmatic ambiguity.
- Investigating MAP-Elites, evolutionary memory, and multi-objective selection to preserve specialist prompts while optimizing accuracy, generalization, and prompt efficiency.

SELECTED PROJECTS

Speak Bridge - Real-Time ASL-to-Speech Prototype **Python, FastAPI, MediaPipe, React**

- Built a multimodal pipeline that streams webcam frames through hand-landmark detection and AI-based interpretation, returning structured sign hypotheses, confidence scores, and visual cues.
- Implemented FastAPI streaming endpoints, live landmark overlays, frontend state handling, and fallbacks for image-processing and API failures.

Clotest Search - ML-Powered Fashion Market Intelligence **React, Node.js, Express, MongoDB**

- Built a multi-marketplace data pipeline that normalizes fashion listings for comparable analysis across brands, categories, conditions, sizes, and providers.
- Developing ML-driven market analysis to estimate fair value, detect pricing anomalies, identify trend signals, and surface resale opportunities for active users in a close-knit fashion community.

Money Magic - AI Financial Wellness Platform **Python, Flask, React, Gemini API**

- Built a data pipeline for CSV ingestion, transaction categorization, recurring-subscription detection, financial summaries, and interactive dashboards.
- Integrated Gemini-powered coaching through modular REST endpoints for dataset upload, analysis, subscriptions, and personalized recommendations.

Actory - Local-First AI Development Orchestrator **Python, FastAPI, SQLite, React**

- Built an agentic development workflow that converts product documents into task queues, compiles coding prompts, captures diffs and logs, verifies changes, and generates repair prompts.
- Designed explicit orchestration states for project setup, task selection, execution, review, retry, and dashboard visibility.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS
AI/ML: LLM evaluation, prompt optimization, evolutionary search, computer vision, MediaPipe, Pandas, Gemini/LLM APIs
Frameworks & Systems: React, FastAPI, Flask, Node.js, Express, MongoDB, SQLite, REST APIs, TailwindCSS, Vite
Developer Tools: Git, GitHub, GitHub Actions, ffmpeg, ngrok

EXPERIENCE & LEADERSHIP

Wegmans - Front End Team Member, Lake Grove, NY **Feb 2025 - Present**

- Deliver accurate service in high-volume operations, manage customer flow, resolve issues, and coordinate with teammates under time pressure.

Moloney's Funeral Homes - Greeter, Long Island, NY **Jan 2024 - May 2026**

- Support families and guests in a sensitive professional environment, communicating service information with discretion and empathy.

Leadership & Honors: Eagle Scout | Bilingual: English and Spanish